

AP EAMCET
Engineering Entrance Exam
25th April 2017- Shift 1

Mathematics

1. Let $\mathbb{N}$ be the set of all natural numbers and $f: \mathbb{N} \rightarrow \mathbb{N}$ be such that $1990 < f(1990) < 2100$ and satisfies the equation

$$x - f(x) = 19 \left[\frac{x}{19} \right] - 90 \left[\frac{f(x)}{90} \right]$$

Where $[y]$ denotes the greatest integer less than or equal to y . Then the number of possible value of $f(1990)$ is

- A.1
 - B.2
 - C.3
 - D.4
2. The function $f: \{1\} \rightarrow \mathbb{R} - \{4\}$ defined by $f(x) = \frac{4x-3}{x-1}$ for $x \in \mathbb{R} - \{1\}$ is
- A. One to but not onto
 - B. Onto but not one-one
 - C. One-one and onto
 - D. Neither one-one nor onto
3. The remainder obtained when 5^{99} is divided by 13 is
- A.6
 - B.7
 - C.8

D.9

4. $\begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} = k(a-b)(b-c)(c-a)$ the $k =$

A. $abc(ab+bc+ca)$

B. $4(ab+bc+ca)(abc)$

C. abc

D. $ab+bc+ca$

5. if P is a square matrix $P^2 = P$ and if I is the unit matrix of the same order as of P then $(P+I)^4 =$

A. $I + 9P$

B. $I + 11P$

C. $I + 13P$

D. $I + 15P$

6. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$

where I is the identity matrix, then $a^2 + b^2 =$

A. 0

B. 2

C. 5

D. 10

7. If $\frac{2+3i\sin\theta}{1-2i\sin\theta}$ is purely imaginary, then $\cos^2\theta =$

A. $\frac{1}{2}$

B. $\frac{3}{2}$

C. $\frac{1}{3}$

D. $\frac{2}{3}$

8. Let A and B represent z_1 and z_2 in the Argand plane and z_1, z_2 be the roots of the equation $Z^2 + pZ + q = 0$, where p, q are complex numbers. If 0 is the origin, $OA = OB$ and $\angle AOB = \alpha$ then $p^2 =$

- A. $2q \cos\left(\frac{\alpha}{2}\right)$
 B. $4q \cos\left(\frac{\alpha}{2}\right)$
 C. $4q \cos^2\left(\frac{\alpha}{2}\right)$
 D. $4q^2 \cos^2\left(\frac{\alpha}{2}\right)$

9. Consider the following statement :

I : If a and b are positive real numbers, then $\sqrt{-a} \times \sqrt{-b} = \sqrt{ab}$

II : The argument of $\frac{1+i\sqrt{3}}{1-i\sqrt{3}}$ is 120°

Then

- A. Only I is true
 B. Only II is true
 C. Both I and II are true
 D. Neither I nor II is true

10. If α is a non – real of the equation $x^6 - 1 = 0$ then $\frac{\alpha^2 + \alpha^3 + \alpha^4 + \alpha^5}{\alpha + 1} =$

- A. α
 B. 1
 C. 0
 D. -1

11. The set of values of $x \in \mathbb{R}$ satisfying the inequality $x^2 - 4x - 21 \leq 0$ is

- A. $[3, 7]$
- B. $[-3, 7]$
- C. $[-7, 3]$
- D. $[-7, -3]$

12. The value of x for which the inequality $\frac{8x^2 + 16x - 51}{(2x - 3)(x + 4)} > 3$ holds, are

- A. $x \geq 4$
- B. $-4 \leq x \leq -3$
- C. $\frac{3}{2} < x < \frac{5}{2}$
- D. $x < -4$ or $x > -3 < x < \frac{3}{2}$

13. The difference of the irrational roots of the equation

$$x^5 - 5x^4 + 9x^3 - 9x^2 + 5x - 1 = 0 \text{ is}$$

- A. $\sqrt{3}$
- B. $2\sqrt{5}$
- C. 3
- D. $\sqrt{5}$

14. For $k > 0$, if $k\sqrt{-1}$ is a root of the equation

$$x^4 - 6x^3 - 16x^2 + 24x - 80 = 0 \text{ then } k^2 =$$

- A. 2
- B. 3
- C. 4
- D. 6

15. The number of ways of awarding 9 scholarships among three students so that each may have 3 scholarships is

- A. 280
- B. 84

C.504

D.1680

16. In the prime factorization of $37! = 2^{\alpha_2} \cdot 3^{\alpha_3} \cdot 5^{\alpha_5} \dots 37^{\alpha_{37}}$ the ratio $\alpha_3 : \alpha_5 =$

A.3:5

B.17:8

C.5:3

D.8:21

17. The number of four digit numbers that can be formed using the digits 1,2,3,4,5,6,7,8,9 which are divisible by 3. When repetition of digits is allowed any number of times, is

A.2187

B.1458

C.6561

D.2916

18. If $x = \frac{2.5}{(2!)3} + \frac{2.5.7}{(3!)3^2} + \frac{2.5.7.9}{(4!)3^3} + \dots$ then $x^2 + 8x + 8$

A.108

B.100

C.27

D.23

19. $3.C_0 + 7.C_1 + 11.C_2 + \dots + (3+4n)C_n =$

A. $(2n+3)2^n$

B. $(2n+1)2^{n-1}$

C. $(2n+3)2^{n-1}$

D. $(2n+1)2^n$

20. If $\frac{x^2 + 5x + 1}{(x+1)(x+2)(x+3)} + \frac{a}{x+1} + \frac{b}{(x+1)(x+2)} + \frac{c}{(x+1)(x+2)(x+3)}$ then the

inverse of the matrix $\begin{bmatrix} a & b \\ c & 1 \end{bmatrix}$ is

A. $\begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix}$

B. $\begin{bmatrix} -1 & 0 \\ 5 & -1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 0 \\ 1 & 5 \end{bmatrix}$

21. $\cos 20^\circ + \cos 30^\circ + \cos 40^\circ =$

A. $1 - 2 \sin 10^\circ \sin 15^\circ \sin 20^\circ$

B. $4 \cos 20^\circ \cos 30^\circ \cos 40^\circ$

C. $4 \cos 10^\circ \cos 15^\circ \cos 20^\circ$

D. $4 \cos 25^\circ \cos 30^\circ \cos 235^\circ$

22. $\cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{14\pi}{15} =$

A. $\frac{9}{16}$

B. $\frac{1}{8}$

C. $\frac{1}{16}$

D. $\frac{1}{24}$

23. If $0 \leq x \leq 2\pi$, then the number of real values of x , which satisfy the equation $\sin x + \sin 2x + \sin 3x + \sin 4x = 0$

A. 9

B. 7

C. 3

D.5

24. The number of solutions of the equation

$$\sin \left[2 \cos^{-1} \left\{ \cot \left(2 \tan^{-1} x \right) \right\} \right] = 0 \text{ is}$$

A.4

B.6

C.8

D. Infinitely many

25. If $a < 1$ and $2 \sinh^{-1} \left(\frac{a}{\sqrt{1-a^2}} \right) = \log \left(\frac{1+x}{1-x} \right)$, then $x =$

A. $2a$

B. $3a$

C. $4a$

D. a

26. If the lengths of two sides of a triangle are the roots of the equation $x^2 - 2\sqrt{3}x + 2 = 0$ and the angle between these sides is $\frac{\pi}{3}$

then the perimeter of the triangle is

A. $2\sqrt{6} + \sqrt{3}$

B. $2\sqrt{6} + 2\sqrt{3}$

C. $\sqrt{6} + 2\sqrt{3}$

D. $\sqrt{6} + \sqrt{3}$

27. If $\triangle ABC$ if $\angle C = 90^\circ$, then $\frac{(\sin^2 A + \sin^2 B)}{(\sin^2 A - \sin^2 B)} \sin(A - B) =$

A. 1

B. $\frac{1}{2}$

C. $\frac{\sqrt{3}}{2}$

D. 0

28. In $\triangle ABC$ if $r_1 = 8, r_2 = 12$ and $r_3 = 24$ then the ordered triple $(a, b, c) =$

- A. (8, 12, 14)
- B. (12, 16, 20)
- C. (16, 12, 20)
- D. (16, 20, 12)

29. A point C with position vector $\frac{3\vec{a} + 4\vec{b} - 5\vec{c}}{3}$ (where $\vec{a}, \vec{b}$ and $\vec{c}$ are non-coplanar vectors) divides the line joining A and B in the ratio 2:1. If the position vector of A is $(\vec{a} - 2\vec{b} + 3\vec{c})$, then the position vector of B is

- A. $2\vec{a} + 3\vec{b} - 4\vec{c}$
- B. $2\vec{a} - 3\vec{b} + 4\vec{c}$
- C. $2\vec{a} + 3\vec{b} + 4\vec{c}$
- D. $\vec{a} + 3\vec{b} - 4\vec{c}$

30. The line passing through the point $2\vec{a} + \vec{b}$ and parallel to the vector $\vec{b} - \vec{c}$ and the plane passing through the point $\vec{a}$ and parallel to the vectors $\vec{b} + \vec{c}$ and $\vec{a} + 2\vec{b} - \vec{c}$ intersect at P. The position vector of P is

- A. $\vec{a} + 3\vec{b}$
- B. $2\vec{a} + 2\vec{b} - \vec{c}$
- C. $\vec{a} + \vec{b} - 2\vec{c}$
- D. $2\vec{a} + \vec{c}$

31. $\vec{a}$ and $\vec{b}$ are not perpendicular to each other,

$\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \cdot \vec{c} = 0$ then $\vec{r} =$

- A. $\vec{a} - \vec{b}$
- B. $\vec{b} - \left(\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{c}} \right) \vec{a}$
- C. $\vec{b} + \left(\frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{c}} \right) \vec{a}$

D. $\bar{a} + \bar{c}$

32. If the two diagonals of a parallelogram are $\bar{i} + 2\bar{j} + 3\bar{k}$ and $-2\bar{i} + \bar{j} - 2\bar{k}$, then the area of the parallelogram in square units is

A. $\frac{3}{2}\sqrt{5}$

B. $5\sqrt{3}$

C. $\sqrt{300}$

D. $3\sqrt{\frac{5}{2}}$

33. The minimum volume of the parallelepiped formed by the vectors $\bar{i} + a\bar{j} + \bar{k}$, $\bar{j} + a\bar{k}$ and $a\bar{i} + \bar{k}$ is

A. $\frac{1}{3\sqrt{3}}$

B. $\frac{1}{\sqrt{3}}$

C. $\frac{3\sqrt{3}-2}{3\sqrt{3}}$

D. $\frac{3\sqrt{3}+2}{3\sqrt{3}}$

34. If $\bar{a} = 2\bar{i} + \bar{j} - 3\bar{k}$, $\bar{b} = \bar{i} - 2\bar{j} + \bar{k}$, $\bar{c} = -\bar{i} + \bar{j} - 4\bar{k}$, $\bar{d} = \bar{i} + \bar{j} + p\bar{k}$, where p is a positive real number $|(\bar{a} \times \bar{b}) \times (\bar{c} \times \bar{d})| = 5\sqrt{114}$, then a value of p is

A. 3

B. 2

C. 1

D. 4

35. The variance of the first ten multiples of 3 is

A. 64.25

B. 54.25

C. 70.75

D.74.25

36. If the mean deviation from the mean of the observations $\{1+k\alpha\}$, where $K=0, 1, 2, \dots, 100$ and $\alpha > 0$ is 255 then $\alpha =$

A.10

B.10.1

C.25

D.25.5

37. Boxes B_1 , B_2 , and B_3 contain balls as given below:

Box	white	Black
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B_1	1	2
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B_2	3	1
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B_3	2	3
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One ball is drawn at random from each box. Then, among the balls drawn, the probability that two are black and one is white. Is

A. $\frac{5}{12}$

B. $\frac{7}{12}$

C. $\frac{9}{12}$

D. $\frac{11}{12}$

38. If A and B are two events of a random experiment such that $P(A) = 0.6$, $P(B) = 0.3$ and $P(A \cap B) = 0.5$ then $P(\bar{B} | \bar{A}) =$

A.0.75

B.0.625

C.0.525

D.0.4

39. A player tosses two coins. He wins Rs. 1 if 1 head appears, Rs. 2 if 2 heads appear. But he loses Rs. 3 if no head appears. The mean of the prized money is
- $\frac{1}{2}$
 - $\frac{1}{3}$
 - $\frac{1}{4}$
 - $\frac{1}{5}$
40. If X is a binomial variate with $n = 7$ and $P(X = 3) = P(X = 4)$ then $P(X = 5) =$
- $21 \cdot \frac{3^2}{4^7}$
 - $21 \cdot \frac{3^5}{4^7}$
 - $21 \cdot \frac{2^5}{3^6}$
 - $\frac{21}{2^7}$
41. If P is a variable point such that the sum of the distances from P to the points $A(2, 2)$ and $B(2, -2)$ is 4, then the locus of P represents
- an ellipse
 - a vertical line
 - segment of a vertical line
 - segment of a horizontal line
42. The point to which the origin is to be shifted to eliminate x and y terms of the equation $4x^2 + 9y^2 - 8x + 36y + 4 = 0$ is
- $(1, 3)$
 - $(-4, 3)$

C. (-1, 2)

D. (1, -2)

43. A line passing through the point P (1, 1) and parallel to the line $x - y = 5$ cuts the line $x + 3y - 2 = 0$ at Q. Then twice the length of the segment PQ is

A. $\sqrt{2} - 1$

B. $2\sqrt{3} - 1$

C. $\sqrt{2}$

D. $\frac{1}{\sqrt{2}}$

44. If the lines $2x + y - 3 = 0$, $3x + 2y - 2 = 0$, $kx - 3y - 23 = 0$ are concurrent, then the roots of the equation $6x^2 - 7x + k = 0$ are

A. $\frac{1}{2}, \frac{2}{3}$

B. 2, 3

C. 3, 4

D. 6, 2

45. A straight line L is perpendicular to the line $5x - y = 1$ and the area of the triangle formed by the line L and the coordinate axes is 5 square units. The equation of the line L can be

A. $x + 5y = \pm 2\sqrt{2}$

B. $x + 5y = \pm 5\sqrt{2}$

C. $x - 5y = 5\sqrt{2}$

D. $-x + 5y = 3\sqrt{2}$

46. The combined equation of the diagonals of the square formed by the two pairs of straight lines given by

$xy + 4x - 3y - 12 = 0$ and $xy - 3x + 4y - 12 = 0$ is

A. $x^2 - 2xy + y^2 + x - y = 0$

B. $x^2 + 2xy + y^2 + x + y = 0$

C. $x^2 - y^2 + x + y = 0$

D. $x^2 - y^2 + x - y = 0$

47. If the pair of lines $2x^2 + 3xy + y^2 = 0$ makes angles θ_1 and θ_2 with the positive direction of the X-axis. then $|\tan(\theta_1 - \theta_2)| =$

A. 1

B. $\frac{1}{3}$

C. $\frac{1}{2}$

D. $\frac{1}{4}$

48. The pair of lines joining the origin to the points of intersection of the line $x - y = 2$ with the curve $5x^2 + 12xy - 8y^2 + 8x - 4y + 12 = 0$ are equally inclined to the pair of lines

A. $x^2 - xy = 0$

B. $xy = 0$

C. $(x - 2)(y - 2) = 0$

D. $xy - y^2 = 4$

49. The locus of the point of intersection of the tangents to the circle $x^2 + y^2 = a^2$ which make complimentary angles with the X-axis is

A. $x^2 - y^2 = 0$

B. $x^2 + y^2 = 0$

C. $xy = 0$

D. $x^2 + y^2 = 2a^2$

50. The length of the chord intercepted by the circle

$$x^2 + y^2 - 4x + 6y - 12 = 0 \text{ on the line } 4x + 3y + 1 = 0 \text{ is}$$

A. 3

B. 4

C. 10

D. 8

51. The inverse point of (1,3) with respect to the circle

$$x^2 + y^2 - 4x - 6y + 9 \text{ is}$$

A. (2,-3)

B. (2,3)

C. (-2,3)

D. (-2,-3)

52. The combined equation of the direct common tangents of the circles

$$x^2 + y^2 - 2x - 2y - 2 = 0 \text{ and } x^2 + y^2 + 4x + 6y + 12 = 0, \text{ is}$$

A. $3x^2 + 10xy + 3y^2 - 4x + 4y - 4 = 0$

B. $75x^2 - 9y^2 - 80xy + 200x + 500y - 2900 = 0$

C. $9x^2 - 16y^2 - 18x + 96y - 135 = 0$

D. $15x^2 - 24xy + 8y^2 - 18x - 8y - 73 = 0$

53. If the circle $x^2 + y^2 = 9$ and $x^2 + y^2 - 8x - 6y + n^2 = 0, n \in \mathbb{Z}$ have exactly two common tangents, then the number of values for n is

A. 8

B. 7

C. 9

D. 4

54. If the angle between the circles

$x^2 - y^2 - 4x - 6y - 3 = 0$, $x^2 + y^2 + 8x - 4y + \lambda = 0$ is 60° , then a value of λ is

- A. -11
- B. 29
- C. 18
- D. -29

55. If tangent lines are drawn from the point $(-1, 2)$ to the parabola $y^2 = 4x$, then the area of the triangle (in sq. units) formed by the chord of contact and the tangents drawn, is

- A. $4\sqrt{2}$
- B. $5\sqrt{2}$
- C. $7\sqrt{2}$
- D. $8\sqrt{2}$

56. If (x_1, y_1) and (x_2, y_2) are the points on the parabola $y^2 = 32x$ each at a focal distance of 10 units, then $2(x_1^2 + x_2^2 + y_1^2 + y_2^2) =$

- A. 68
- B. 136
- C. 272
- D. 8

57. The coordinates of a point, in the parametric form, on the ellipse whose foci are $(-1, 0)$ and $(7, 0)$ and eccentricity $\frac{1}{2}$.

Are

- A. $(8\cos\theta, 4\sqrt{3}\sin\theta)$
- B. $(3+8\cos\theta, 4\sqrt{3}\sin\theta)$
- C. $(3+4\sqrt{3}\cos\theta, 8\sin\theta)$
- D. $(3+4\cos\theta, 2\sqrt{3}\sin\theta)$

58. If the normal at one end of a latus rectum of the ellipse

$$\frac{x^2}{32} + \frac{y^2}{b^2} = 1 \text{ passes through one}$$

end of the minor axis, then $\frac{e^4}{1-e^2} = \underline{\hspace{2cm}}$ (Here e is the eccentricity of the ellipse)

A. $\frac{1}{2}$

B. 1

C. $\frac{2}{3}$

D. $\frac{3}{2}$

59. The locus of the mid points of the chords of the circle $x^2 + y^2 = 16$. which are tangents to the hyperbola $9x^2 - 16y^2 = 144$. is

A. $8x^2 - 9y^2 = (x^2 + y^2)^2$

B. $16x^2 - 9y^2 = (x^2 + y^2)^2$

C. $9x^2 - 14y^2 = (x^2 + 2y^2)^2$

D. $3x^2 + 49y^2 = (x^2 + 2y^2)^2$

60. The coordinates of the point at which X Z - plane divides the line joining A (-2, 3, 4) and B (1, 2, 3) is

A. (7, 0, -1)

B. (7, 0, 1)

C. (-7, 0, 1)

D. (-7 0, -1)

61. Direction ratios of normal to a plane passing through (1, 0, 0) and (0, 1, 0) which makes an angle of $\frac{\pi}{4}$ with the plane $x + y - 3 = 0$ are

A. $(\sqrt{2}, 1, 1)$

B. $(1, \sqrt{2}, 1)$

C. $(1, 1, \sqrt{2})$

D. $(1, 1, 2)$

62. A plane π is passing through $(2, 0, 1)$ and $(3, -3, 4)$ and is perpendicular to $x - 2y + z = 6$. Among the following, the plane that is perpendicular to the plane π is

A. $3x + 2y + z = 7$

B. $3x - 2y + z = 7$

C. $x - 2y - z + 1 = 0$

D. $3x - 2y - z = 7$

63. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = [x] + \sqrt{x - [x]}$ for $x \in \mathbb{R}$, where $[x]$ denotes the greatest integer function. Then the set of points at which f is continuous is

A. $\mathbb{R}^+$

B. $\mathbb{R}$

C. $\mathbb{R} - \mathbb{Z}$

D. $\{1, 2, 3, \dots\}$

64. If α, β are the roots of the equation

$$ax^2 + bx + c = 0. \text{ then } \lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2} =$$

A. $a^2(\alpha - \beta)^2$

B. $4a^2(\alpha - \beta)^2$

C. $\frac{a^2}{2}(\alpha - \beta)^2$

D. $2a^2(\alpha - \beta)^2$

65. If $f(x) = \tan^{-1} \left[\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right]$ for $0 > |x| < 1$ then $f'(x) =$

A. $\frac{x}{\sqrt{1-x^2}}$

B. $\frac{-x}{\sqrt{1-x^2}}$

C. $\frac{x}{\sqrt{1-x^4}}$

D. $\frac{-x}{\sqrt{1-x^4}}$

66. If $y = a \cos(\log x) + b \sin(\log x)$ then $x^2 y_2 + x y_1 =$

A. 0

B. Y

C. 2y

D. -y

67. If $x = a(\cos t + t \sin t)$, $y = a(\sin t - t \cos t)$ then $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} =$

A. a

B. at

C. $a^2 t^2$

D. $a^2 t$

68. if $ab \neq 0$, then the equation of the tangent at (a,b) to the curve

$$\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2 \text{ is}$$

A. $\frac{x}{a} + \frac{y}{b} = 1$

B. $ax + by = 1$

C. $\frac{x}{a} + \frac{y}{b} = 2$

D. $ax + by = 2n$

69. A particle is moving along a line according to the law $S = t^3 - 3t^2 + 4t - 2$ where S is measured in meters and t is measured in seconds. Then the velocity (in mt/sec) of the particle when its acceleration is zero is

A. 2

B. $\frac{1}{4}$

C. $\frac{17}{4}$

D. 1

70. The greatest value of the real-valued function

$f(x) = (x+1)^{\frac{1}{3}} - (x-1)^{\frac{1}{3}}$ on $[0,1]$ is

A. $f(x) = (x+1)^{\frac{1}{3}} - (x-1)^{\frac{1}{3}}$ on $[0,1]$

B. 1

C. 2

D. 3

71. The constant c of Rolle's theorem for the function

$f(x) = (x-1)^3 (x-2)^5$ in $[1,2]$ is

A. $\frac{3}{2}$

B. $\frac{11}{6}$

C. $\frac{13}{8}$

D. $\frac{11}{8}$

72. $\int \frac{1}{x\sqrt{x^6+1}} dx =$

A. $\frac{1}{3} \sinh^{-1} \left(\frac{1}{x^3} \right) + C$

B. $-\frac{1}{3} \sinh^{-1} (x^3) + C$

C. $\frac{-1}{3} \sinh^{-1} \left(\frac{1}{x^3} \right) + C$

D. $3 \sinh^{-1} \left(\frac{1}{x^3} \right) + C$

73. $\int x^2 \sin x \cos x dx =$

A. $-\frac{x^2 \cos 2x}{2} + \frac{x \sin 2x}{4} + \frac{\cos 2x}{8} + c$

B. $\frac{(1-2x)^2}{2} \cos 2x + x \sin 2x + c$

C. $\frac{(1-2x)^2}{8} \cos 2x + \frac{x}{4} \sin 2x + c$

D. $\frac{(1-2x)^2}{4} \cos 2x + \frac{x}{2} \sin 2x + c$

74. $\int \frac{dx}{x^{2/3} (1+x^{2/3})} =$

A. $3 \sin^{-1} \left(x^{1/3} \right) + c$

B. $3 \cos^{-1} \left(x^{1/3} \right) + c$

C. $3 \tan^{-1} \left(x^{1/3} \right) + c$

D. $3 \sec^{-1} \left(x^{1/3} \right) + c$

75. $I_n = \int \frac{t^n}{1+t^2} dt, (n=1, 2, 3, \dots) \Rightarrow I_6 + I_4 =$

A. $\frac{1}{5} t^5 + c$

B. $\frac{1}{7} t^7 + c$

C. $\frac{1}{4} t^4 + c$

D. $\frac{1}{3} t^3 + c$

76. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{r}{\sqrt{n^2 + r^2}} =$

A. $1 + \sqrt{5}$

B. $-1 + \sqrt{5}$

C. $-1 + \sqrt{2}$

D. $1 + \sqrt{2}$

77. $\int_0^{\pi/2} \sin^8 x \, dx =$

A. $\frac{15\pi}{256}$

B. $\frac{25\pi}{256}$

C. $\frac{35\pi}{256}$

D. $\frac{35}{256}$

78. The area (in sq. units) of the region bounded by the parabola $y^2 = 4x$ and $y^2 = 4(4 - x)$

A. $\frac{7}{4} - \sqrt{2}$

B. $2(\sqrt{2} - 1)$

C. $\frac{32\sqrt{2}}{3}$

D. $\frac{8}{3}$

79. The differential equation of the family of curves given by $ae^{2x} + be^{5x}$ where a and b are parameters is

A. $\frac{d^2 y}{dx^2} + 7 \frac{dy}{dx} - 10y = 0$

B. $\frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} + 10y = 0$

C. $\frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} - 12y = 0$

D. $\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 7y = 0$

80. The equation of the curve passing through $\left(\frac{\pi}{6}, 0\right)$ and satisfying differential equation $(e^y + 1)\cos x dx + e^y \sin x dy = 0$

A. $y = \log_e (\cos ex - 1)$

B. $y = \log_e (\cos ex + 1)$

C. $y = \log_e (y + 1)$

D. $y = \log_e (y - 1)$